



Kishoreganj University
3rd Year 2nd Semester B. Sc. (Engg.) Final Examination-2024
Department of Computer Science and Engineering
CSE 3205: Web and Mobile Application Development (3 Credits)

Time: 3 Hours

Full Marks: 70

Instructions

- Figures shown in the right margin indicate full marks.
- Answer 5 out of 7 questions.

1. a. Describe the behavior of Block-level and Inline elements using suitable examples. Also, describe a situation where the visual size of an <h6> tag appears larger than an <h1> tag and the process required to achieve this outcome. **04**
- b. The ICT Cell of Kishoreganj University is improving the design of a student profile webpage. During testing, multiple CSS rules are applied to the same elements, causing style conflicts. Illustrate and analyze the given CSS and HTML code to determine the final background color displayed in each element. **04**

Code Snippet:

```
<style>
  div { background-color: grey; }
  .container { background-color: yellow !important; }
  #profile-box { background-color: blue; }

  header { background-color: pink; }
  .nav-header { background-color: green; }
  #main-header { background-color: orange; }

  section p { background-color: cyan; }
  .text-box p { background-color: brown; }

  span { background-color: purple; }
  .status-text { background-color: silver; }
</style>

<div id="profile-box" class="container">Profile Area</div>

<header id="main-header" class="nav-header" style="background-color: red;">
  Welcome to KiU
</header>

<section class="text-box">
  <p>Student Information</p>
</section>

<span class="status-text" style="background-color: white;">Active</span>
```

Determine the final Background Color for the following four parts:

- The div element with the text "Profile Area".
 - The header element with the text "Welcome to KiU".
 - The p element with the text "Student Information".
 - The span element with the text "Active".
- c. Consider the following snippet is developed using the Bootstrap 5 CSS framework. Assume all necessary CDNs are properly imported. **06**

Code Snippet:

```
<div class="container border">
  <div class="row">
    <div class="col-12 col-md-3 border">University Logo</div>
    <div class="col-12 col-md-9 border">
      <div class="row">
        <div class="col-12 col-md-6 border">
          <div class="row">
            <div class="col-6 border">Semester</div>
            <div class="col-6 border">Credit</div>
          </div>
        </div>
        <div class="col-12 col-md-6 border">
          <div class="row">
            <div class="col-4 border">GPA</div>
            <div class="col-8 border">Attendance</div>
          </div>
        </div>
      </div>
    </div>
  </div>
</div>
```

Bootstrap Breakpoints Reference is given in the following table:

Breakpoint	Class prefix	Dimensions
Extra Small	.col-	< 576px
Small	.col-sm-	≥ 576px
Medium	.col-md-	≥ 768px
Large	.col-lg-	≥ 992px

Now

- A. Illustrate the visual layout of the code for Screen Width = 500px.
- B. Illustrate the visual layout of the code for Screen Width = 1000px.

- 2
- a. A food delivery app must display:

customer location

restaurant location

delivery rider live location

route estimation

04

However, the rider location updates every second, causing battery drain. Design an optimized location tracking system. Validate how your approach reduces battery usage while maintaining accuracy.

- b. A learning app allows users to download video lectures and watch offline. Storage usage becomes too high and phones run out of space. Construct a multimedia storage management policy. Justify how your approach balances storage efficiency and video quality.

c. A video conferencing iOS app uses high memory when switching between screens. Users experience frequent app termination. Optimize the memory usage strategy. Propose solutions involving weak references, cleanup methods, and resource unloading.

a. Illustrate the structure of JSON by comparing its key-value pair format with a row in a traditional SQL table, and explain how an Android application converts a JSON snapshot from Firebase into a usable class-object inside the code for further processing.
- 04
- 06
- 04
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- 5/5
- b. Explain why an Android application cannot directly fetch data at page load like a PHP-MySQL web page, and explain why event-based mechanisms such as OnClickListener or Firebase event listeners are required to retrieve data from Firebase Realtime Database in a mobile environment. **04**
- c. Consider a "Blood Donor Tracker" android application for a local hospital using Firebase Realtime Database. You need to create a Java class that represents a Donor. Each donor must have a name, bloodGroup, and phoneNumber in String format. Now you have to design a Donor class in a way that: **06**
- Firebase can automatically reconstruct the object when it fetches data from the cloud.
 - The developer can create a new donor object with all their information at once while the app is running (A constructor that initializes all class variables).
- ✓ a. Evaluate why JSON is more popular than XML in modern web services. Also mention situations where XML may still be preferred. **04**

- b. Consider the two tables below. Determine the exact output of the PHP script. Show each step of the join and calculation. **04**

Table: *products*

<i>p_id</i>	<i>p_name</i>	<i>base_price</i>	<i>cat_id</i>
101	Laptop	50000	1
102	Mouse	1000	2
103	Monitor	15000	1

Table: *categories*

<i>cat_id</i>	<i>cat_name</i>	<i>tax_rate</i>
1	Electronics	10
2	Accessories	5

PHP Code:

```
<?php
$sql = "SELECT p.p_name, p.base_price, c.tax_rate
        FROM products p
        JOIN categories c ON p.cat_id = c.cat_id
        WHERE p.base_price > 5000";

$result = mysqli_query($conn, $sql);
$total_tax = 0;

while ($row = mysqli_fetch_assoc($result)) {
    $calculated_tax = ($row['base_price'] * $row['tax_rate']) / 100;
    $total_tax += $calculated_tax;
    echo "Item: " . $row['p_name'] . " | Tax: " . $calculated_tax . "<br>";
}
echo "Total Tax: " . $total_tax;
?>
```

- c. You have been recently appointed at the ICT Cell of Kishoreganj University. On your very first day, you faced a serious challenge: some students are accidentally logging into other students' portals. The developer made a mistake and some students might have been entered twice in the database. To solve this problem, you have been assigned to write a PHP script that will identify these duplicate IDs. You need to process the student list from a table called *new_admissions*. **06**

Database Connection Info:

- Database Name: kiu_db_students
- Username: admin_kiu_ict
- Password: p@ssworD123

admin '1' --
OR '1' = '1'

OR 1=1#
OR 1=1--

Note that, The database and your PHP script are on the same server.

Your Tasks:

- **Database Connection:** Write the PHP code to connect to the database using the given credentials.
- **Fetch Data:** Write a SQL query in your script to get the student_id and email from the new_admissions table.
- **Find Duplicates using Associative Array:**
 - Declare an empty Associative Array named \$identity_tracker.
 - Use a loop to check each student record.
 - Hint: Use the student_id as the Key of \$identity_tracker. For every student, check if their student_id is already present in the \$identity_tracker.
- **Final Output:** After the loop, print the total duplicate students ids with their email so that you can email them later when their problem is solved.

- 5 a. Explain how an Intent is created and used to navigate between activities in mobile application development, and show how data is passed using putExtra() and retrieved in the target activity using getStringExtra(). 04
- b. A beginner Android developer is confused after switching between "Android View" and "Project View" in Android Studio. Explain the practical differences between these two views and describe a situation where "Project View" would be required. 04
- c. You are asked to build a mini product listing page for a e-commerce app that includes: 06
- product cards (image, name, price)
 - add-to-cart button
 - filter by price (low to high, high to low)
 - responsive UI

Create the full approach (HTML structure + CSS layout + JavaScript logic). Then evaluate how your design supports scalability for large product data.

- 6 a. Explain why using the GET method is not suitable for passing login information and what problems may occur. 04
- b. An e-commerce website allows unlimited login attempts without any restriction in password format. After some time, many user accounts get hacked. Explain what kind of attack may have happened and describe at least two ways to prevent this problem. 04
- c. The following PHP code for user login has been written in a web application: 06

PHP Code:

```
$username = $_POST['username'];  
$password = $_POST['password'];  
  
$query = "SELECT * FROM users WHERE username='$username' AND  
password='$password'";  
  
$result = mysqli_query($conn, $query);
```

- A. Write at least three different malicious inputs that can be used to perform SQL Injection in this login system.
- B. Explain why this code is vulnerable.
- C. Describe how you can fix this problem.

- 7 a. In an Android app layout, a developer sets layout_width="0dp" for a TextView inside a LinearLayout but still sees it properly displayed. Explain why 0dp is used in this case and how weight distribution works. 04

b. Describe the visual difference between `android:layout_width="match_parent"` and `android:layout_width="wrap_content"` for a Button. State the purpose of using the `android:id` attribute in an XML element and explain how it helps the backend code (Java) to interact with that specific element. **04**

c. Design a full architecture for an Android e-commerce app that includes: **06**

- UI activities
- intents navigation
- resource handling
- local storage

Explain how these components interact.